01_antonelli_vs_russell

May 13, 2026

1 Antonelli vs Russell — 2026 Qualifying, First Four Rounds

I wanted to see where Kimi Antonelli's 2026 season at Mercedes actually stands vs his teammate George Russell. This notebook compares both Mercedes drivers' fastest qualifying laps across the first four rounds of 2026, broken down by track segment. See the project README for full method and limitations.

1.1 Method (briefly)

For each race: load qualifying, take each driver's fastest valid lap, and segment the track from FastF1's `circuit_info.corners` (corners within 250 m are grouped, straights are filled in between). Telemetry — including FastF1's per-sample `Time` channel — is resampled onto a uniform 5 m distance grid. Per-segment time is read directly from that resampled `Time` channel as `Time[last_step] - Time[first_step]` inside the segment. Per-segment delta is `time(Russell) - time(Antonelli)`, so **positive = Antonelli faster**.

> An earlier version integrated `step / speed` per grid point. On 2 of 4 races the accumulated residual exceeded the 0.1 s sanity-check threshold (see the next section), so the analysis switched to reading FastF1's sample times directly. Residuals are now 0.1 s on all four races.

Each segment is classified into one of four categories by its minimum speed: slow corner (<130 kph), medium corner (130–200 kph), fast corner (>200 kph), or straight.

```
core            INFO     Loading data for Australian Grand Prix - Qualifying
[v3.8.1]
req             INFO     Using cached data for session_info
req             INFO     Using cached data for driver_info
req             INFO     Using cached data for session_status_data
req             INFO     Using cached data for track_status_data
req             INFO     Using cached data for _extended_timing_data
req             INFO     Using cached data for timing_app_data
core            INFO     Processing timing data…
core         WARNING     No lap data for driver 18
core         WARNING     No lap data for driver 55
```

```
core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 18)

core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 55)

req            INFO     Using cached data for car_data

req            INFO     Using cached data for position_data

req            INFO     Using cached data for weather_data

req            INFO     Using cached data for race_control_messages

core           INFO     Finished loading data for 22 drivers: ['63', '12', '6',
'16', '81', '1', '44', '30', '41', '5', '27', '87', '31', '10', '23', '43',
'14', '11', '77', '18', '3', '55']

core           INFO     Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

req            INFO     Using cached data for session_status_data

req            INFO     Using cached data for track_status_data

req            INFO     Using cached data for _extended_timing_data

req            INFO     Using cached data for timing_app_data

core           INFO     Processing timing data…

req            INFO     Using cached data for car_data

req            INFO     Using cached data for position_data

req            INFO     Using cached data for weather_data

req            INFO     Using cached data for race_control_messages

core           INFO     Finished loading data for 22 drivers: ['12', '63', '44',
'16', '81', '1', '10', '3', '6', '87', '27', '43', '31', '30', '41', '5', '55',
'23', '14', '77', '18', '11']

core           INFO     Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

req            INFO     Using cached data for session_status_data

req            INFO     Using cached data for track_status_data

req            INFO     Using cached data for _extended_timing_data

req            INFO     Using cached data for timing_app_data
```

```
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 22 drivers: ['12', '63', '81',
'16', '1', '44', '10', '6', '5', '41', '3', '31', '27', '30', '43', '55', '23',
'87', '11', '77', '14', '18']
core           INFO     Loading data for Miami Grand Prix - Qualifying [v3.8.1]
req            INFO     Using cached data for session_info
req            INFO     Using cached data for driver_info
req            INFO     Using cached data for session_status_data
req            INFO     Using cached data for track_status_data
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 22 drivers: ['12', '3', '16',
'1', '63', '44', '81', '43', '10', '27', '30', '87', '55', '31', '23', '41',
'14', '18', '77', '11', '5', '6']
```

```
   Round                   Race  ANT lap (s)  RUS lap (s)  Δ lap (s)  ANT Q  \
0      1  Australian Grand Prix       78.811       78.518     -0.293      3
1      2     Chinese Grand Prix       92.064       92.286      0.222      3
2      3    Japanese Grand Prix       88.778       89.076      0.298      3
3      4       Miami Grand Prix       87.798       88.197      0.399      3

   RUS Q  Q mismatch
0      3       False
1      3       False
2      3       False
3      3       False
```

No Q-session mismatches across the 4 races.

1.2 Sensor quality

FastF1 telemetry doesn't carry an explicit quality flag, but a frozen `Speed` sensor produces a clear signature: only a handful of unique speed values across many samples within one segment. Any segment that fails this check is excluded from the headline category analysis below — the per-segment time delta is also unreliable in those segments because cumulative `Distance` is integrated from the bad speed.

```
Flagged 5 segment(s) where a driver's speed sensor looks frozen:

                  race segment      kind       category  min_speed_kph  \
0  Japanese Grand Prix      S7  straight       straight          177.2
1  Japanese Grand Prix      C7    corner  medium_corner          189.0
2  Japanese Grand Prix      S8  straight       straight          189.0
3  Japanese Grand Prix      C8    corner  medium_corner          189.0
4  Japanese Grand Prix      S9  straight       straight           97.0

   delta_s_pre_filter
0              -3.555
1              -0.766
2              -1.663
3              -2.416
4               8.712
```

1.3 Sanity check

The segmentation logic is only useful if the segment deltas sum back to the actual lap-time delta. The table below verifies this on real telemetry: any `|Δ|` above 0.1 s would indicate a bug.

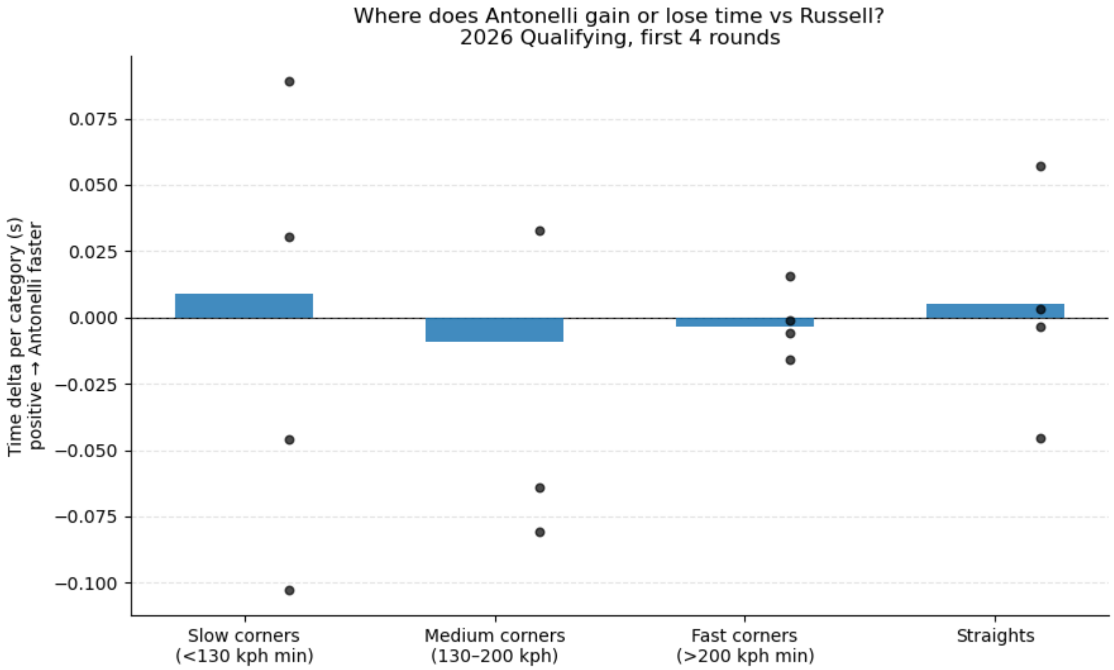
	Race	Σ segment Δ (s)	Lap-time Δ (s)	|residual| (s)
0	Australian Grand Prix	-0.3838	-0.293	0.09084
1	Chinese Grand Prix	0.2352	0.222	0.01318
2	Japanese Grand Prix	0.2758	0.298	0.02216
3	Miami Grand Prix	0.3758	0.399	0.02316

category	fast_corner	medium_corner	slow_corner	straight
event_name				
Australian Grand Prix	3	1	3	8
Chinese Grand Prix	2	1	5	9
Japanese Grand Prix	2	5	1	9
Miami Grand Prix	2	0	4	7

1.4 Headline finding — where does the time go?

Each bar is the mean per-category delta across all four races. Black dots show each individual race's per-category mean — they tell you whether the average reflects a consistent pattern or a noisy one.

```
Headline chart uses 57 segments after excluding 5 with sensor-freeze flags.
```




```
Per-category means after filtering (s/lap, positive = ANT faster):
category
fast_corner      -0.0034
medium_corner    -0.0093
slow_corner       0.0088
straight          0.0051
```

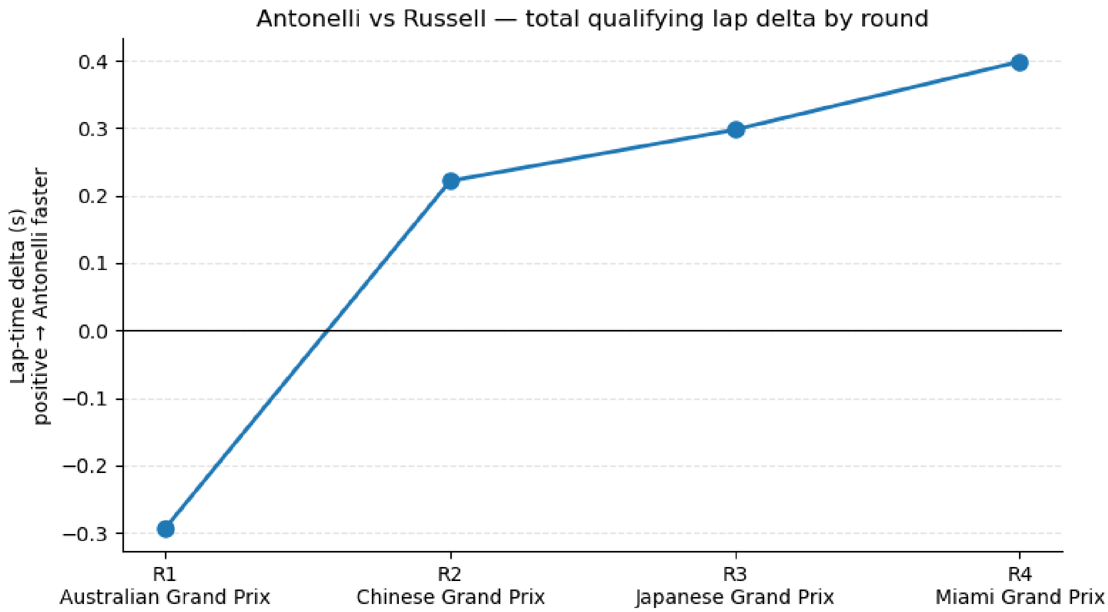
The segment-level picture, after excluding five Japan segments where Antonelli's speed sensor was frozen for ~1300 m of the lap, is quieter than I expected:

- **Straights:** essentially tied (**+0.005 s/lap**).
- **Slow corners (<130 kph):** essentially tied (**+0.009 s/lap**).
- **Fast corners (>200 kph):** essentially tied (**−0.003 s/lap**).
- **Medium corners (130–200 kph):** essentially tied (**−0.009 s/lap**).

Earlier versions of this analysis (before the sensor-quality filter was tightened) reported large per-category deltas — "+0.17 s/lap on straights" and "−0.46 s/lap in medium corners." Both numbers turned out to be driven almost entirely by five segments at Japan where the speed sensor was stuck near 189 kph. Once those segments are removed, no single category emerges as the source of Antonelli's lap-time advantage; the +0.16 s/lap mean over four races is spread across the whole lap rather than concentrated in any one phase.

The cleaner read: at four races and with one of them partially compromised, the trajectory finding (next section) is what the data actually supports.

1.5 Trajectory — is the gap closing?



Across four rounds, the lap-time gap has gone from **−0.29 s** (Russell faster by 0.29 s) at Australia to **+0.40 s** (Antonelli faster by 0.40 s) at Miami — a monotone trend in Antonelli's favour covering **0.69 s across four races**. Four data points is too few to project a trajectory with confidence, but the direction is striking: Antonelli was behind at Round 1 and ahead on every subsequent round, with the margin growing each time. Australia looks like the outlier round, not the norm.

1.6 Where on the track is each driver gaining time?

The category breakdown averages across each segment, but it can hide where exactly within a segment the time is found. Below, each circuit is shown as a track map coloured by the **local slope of the cumulative time delta** — the rate at which the gap is changing as the lap unfolds.

- **Blue** = Antonelli is gaining on Russell at that part of the track.
- **Red** = Russell is gaining on Antonelli.
- Corner numbers (from FastF1's `circuit_info`) are overlaid for orientation.

```
core            INFO     Loading data for Australian Grand Prix - Qualifying
[v3.8.1]

req             INFO     Using cached data for session_info

req             INFO     Using cached data for driver_info

req             INFO     Using cached data for session_status_data

req             INFO     Using cached data for track_status_data

req             INFO     Using cached data for _extended_timing_data

req             INFO     Using cached data for timing_app_data
```

```
core           INFO     Processing timing data…

core        WARNING     No lap data for driver 18

core        WARNING     No lap data for driver 55

core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 18)

core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 55)

req            INFO     Using cached data for car_data

req            INFO     Using cached data for position_data

req            INFO     Using cached data for weather_data

req            INFO     Using cached data for race_control_messages

core           INFO     Finished loading data for 22 drivers: ['63', '12', '6',
'16', '81', '1', '44', '30', '41', '5', '27', '87', '31', '10', '23', '43',
'14', '11', '77', '18', '3', '55']

core           INFO     Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

req            INFO     Using cached data for session_status_data

req            INFO     Using cached data for track_status_data

req            INFO     Using cached data for _extended_timing_data

req            INFO     Using cached data for timing_app_data

core           INFO     Processing timing data…

req            INFO     Using cached data for car_data

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req            INFO     Using cached data for weather_data

req            INFO     Using cached data for race_control_messages

core           INFO     Finished loading data for 22 drivers: ['12', '63', '44',
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'23', '14', '77', '18', '11']

core           INFO     Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

req            INFO     Using cached data for session_status_data
```

```
req             INFO      Using cached data for track_status_data
req             INFO      Using cached data for _extended_timing_data
req             INFO      Using cached data for timing_app_data
core            INFO      Processing timing data…
req             INFO      Using cached data for car_data
req             INFO      Using cached data for position_data
req             INFO      Using cached data for weather_data
req             INFO      Using cached data for race_control_messages
core            INFO      Finished loading data for 22 drivers: ['12', '63', '81',
'16', '1', '44', '10', '6', '5', '41', '3', '31', '27', '30', '43', '55', '23',
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core            INFO      Loading data for Miami Grand Prix - Qualifying [v3.8.1]
req             INFO      Using cached data for session_info
req             INFO      Using cached data for driver_info
req             INFO      Using cached data for session_status_data
req             INFO      Using cached data for track_status_data
req             INFO      Using cached data for _extended_timing_data
req             INFO      Using cached data for timing_app_data
core            INFO      Processing timing data…
req             INFO      Using cached data for car_data
req             INFO      Using cached data for position_data
req             INFO      Using cached data for weather_data
req             INFO      Using cached data for race_control_messages
core            INFO      Finished loading data for 22 drivers: ['12', '3', '16',
'1', '63', '44', '81', '43', '10', '27', '30', '87', '55', '31', '23', '41',
'14', '18', '77', '11', '5', '6']
```
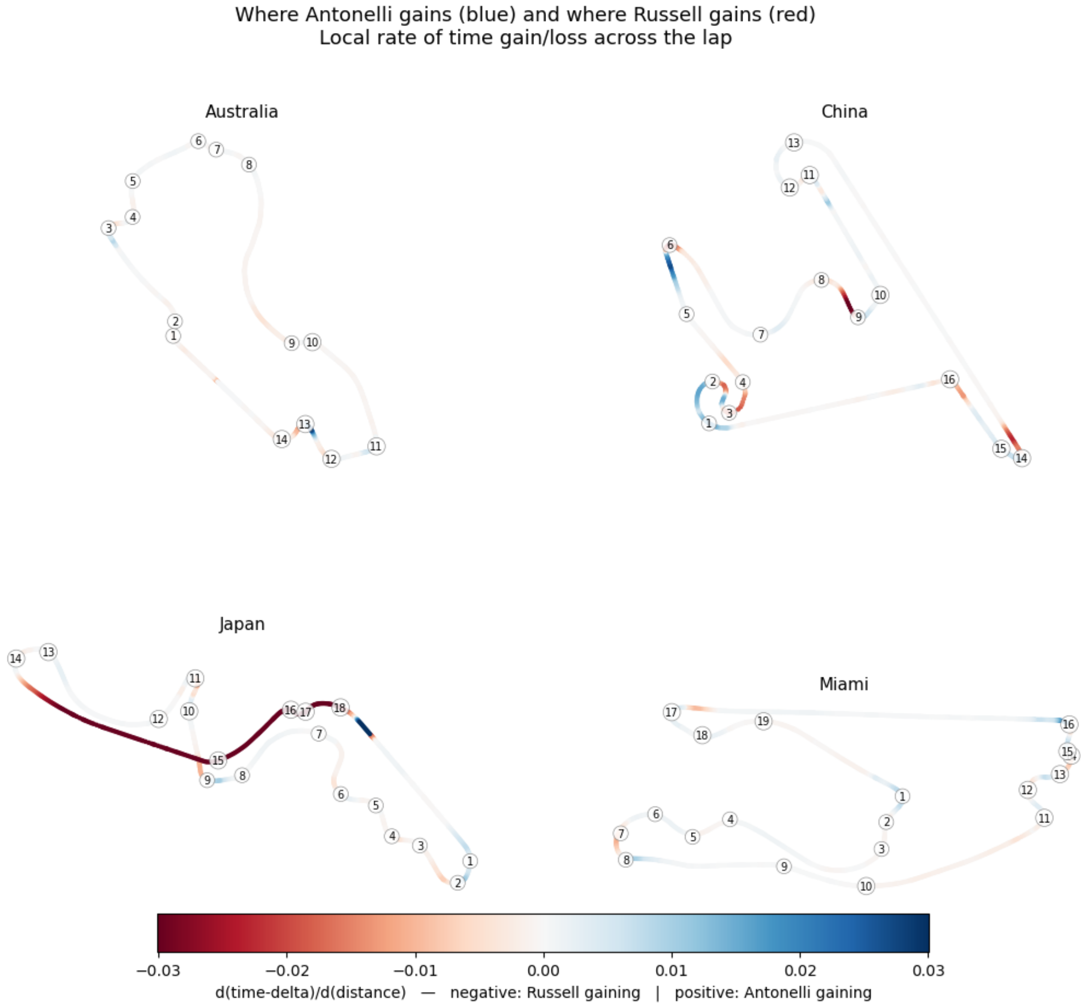



1.7 How is he winning?

We know Antonelli is faster — at every track, every round. But how, specifically? The segment-level breakdown (above) turned out flat: straights, fast corners, slow corners, medium corners all essentially tied. So the advantage has to be hiding in something that segment-time averaging blurs over.

To find it, I looked at two specific signals at each corner on the lap. **Apex speed**: how much speed each driver carries through the deepest point of the corner. **Brake and throttle timing**: how late each driver brakes going in, and how soon they get back on throttle coming out.

1.7.1 Apex speed isn't where he's winning

```
core            INFO     Loading data for Australian Grand Prix - Qualifying
[v3.8.1]

req             INFO     Using cached data for session_info
```

```
req            INFO     Using cached data for driver_info
req            INFO     Using cached data for session_status_data
req            INFO     Using cached data for track_status_data
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
core        WARNING     No lap data for driver 18
core        WARNING     No lap data for driver 55
core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 18)
core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 55)
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 22 drivers: ['63', '12', '6',
'16', '81', '1', '44', '30', '41', '5', '27', '87', '31', '10', '23', '43',
'14', '11', '77', '18', '3', '55']
core           INFO     Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]
req            INFO     Using cached data for session_info
req            INFO     Using cached data for driver_info
req            INFO     Using cached data for session_status_data
req            INFO     Using cached data for track_status_data
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
```

```
core           INFO     Finished loading data for 22 drivers: ['12', '63', '44',
'16', '81', '1', '10', '3', '6', '87', '27', '43', '31', '30', '41', '5', '55',
'23', '14', '77', '18', '11']
core           INFO     Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]
req            INFO     Using cached data for session_info
req            INFO     Using cached data for driver_info
req            INFO     Using cached data for session_status_data
req            INFO     Using cached data for track_status_data
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 22 drivers: ['12', '63', '81',
'16', '1', '44', '10', '6', '5', '41', '3', '31', '27', '30', '43', '55', '23',
'87', '11', '77', '14', '18']
core           INFO     Loading data for Miami Grand Prix - Qualifying [v3.8.1]
req            INFO     Using cached data for session_info
req            INFO     Using cached data for driver_info
req            INFO     Using cached data for session_status_data
req            INFO     Using cached data for track_status_data
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 22 drivers: ['12', '3', '16',
'1', '63', '44', '81', '43', '10', '27', '30', '87', '55', '31', '23', '41',
'14', '18', '77', '11', '5', '6']
```

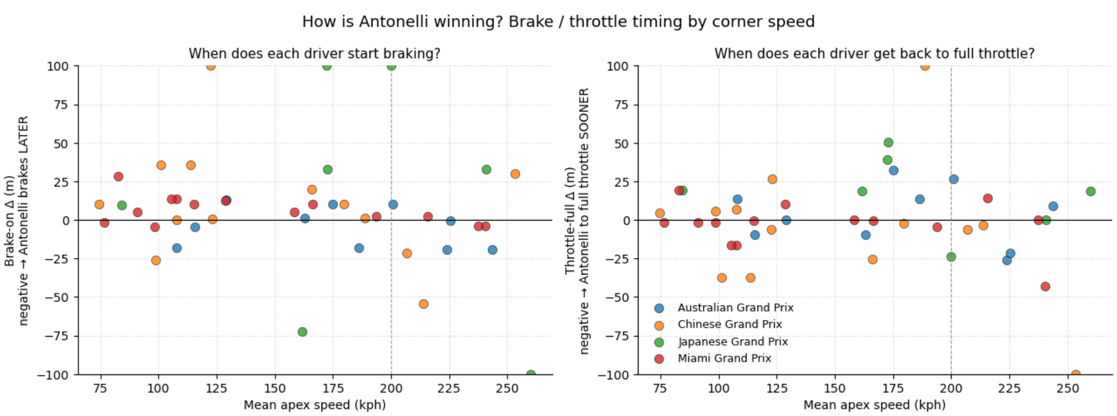
```
Apex-speed delta by corner-speed bucket  (positive = Antonelli faster at the
apex):

                  mean ANT-RUS (kph)   n
speed_bucket
very slow (<100)               -2.93   7
slow (100-150)                  1.21  12
medium (150-200)               -0.58  14
fast (>=200)                    2.76  30
```

At slow and medium corners — the ones with a real apex where commitment matters — Antonelli is tied or slightly behind Russell. He's not carrying noticeably more speed through the deepest part of the corner. The narrow positive at fast corners is real but small, and many of those "corners" in FastF1's circuit info are really high-speed kinks where the apex notion doesn't apply cleanly.

So raw apex commitment isn't the answer. The advantage has to come from somewhere else in the corner cycle.

1.7.2 He brakes later and gets back on throttle sooner — but only at fast corners

```
Brake-on Δ (negative = Antonelli brakes LATER)
Throttle-full Δ (negative = Antonelli back on full throttle SOONER)

                 brake_on_delta       throttle_full_delta
                           mean count                mean count
speed_bucket
very slow (<100)            3.1     7                 6.3     7
slow (100-150)             21.6    12                -5.5    12
medium (150-200)           18.9    13                19.4    13
fast (>=200)              -20.9    12               -23.5    12
```



There it is. At fast corners — and only at fast corners — Antonelli brakes about 21 m later than Russell on average AND gets back to full throttle about 23 m sooner. That's the late-brake / early-throttle commitment signature you'd expect from a young driver willing to push the entry and exit of high-speed turns.

At slower corners his approach is the opposite: he brakes *earlier* and gets back on the gas *later* than Russell. It's almost like two different driving philosophies depending on speed range.

So the mechanism is: he wins at fast corners through commitment in the braking and traction zones, and gives a bit of time back at slow corners. The fast-corner gain wins out on net.

A real caveat: 44 corners across 4 races, only n=12 in the fast bucket, and a handful of dramatic outliers (Japan T8 alone has Russell braking 200 m earlier than Antonelli at a 260-kph kink). The fast-corner signature is consistent and large enough to feel real, but I'd want more clean races before calling it final.

1.8 Year-over-year: rookie vs sophomore

The within-2026 segment-level breakdown collapsed under the data-quality filter, but the lap-level comparison against 2025 — Antonelli's actual rookie season at Mercedes — is far more striking. The cell below pulls the same 4 races from 2025 and plots both years on one chart.

At each of these tracks, Antonelli has gained between 0.3 and 0.7 s on Russell year-over-year. He's not just continuing his rookie-year closing arc; he's compressing it: he was almost a second behind Russell at his Australia debut and only just crossed ahead by Miami (R6) — in 2026 he started at −0.29 s and was ahead from R2 onwards.

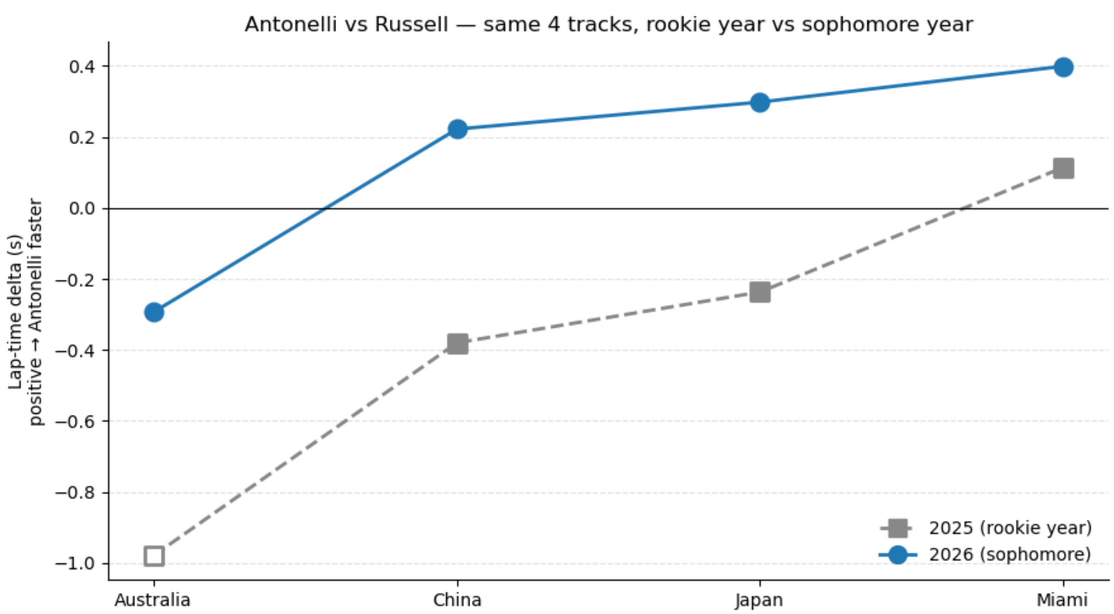
```
core            INFO     Loading data for Australian Grand Prix - Qualifying
[v3.8.1]

req             INFO     Using cached data for session_info

req             INFO     Using cached data for driver_info

req             INFO     Using cached data for session_status_data

req             INFO     Using cached data for track_status_data

req             INFO     Using cached data for _extended_timing_data

req             INFO     Using cached data for timing_app_data

core            INFO     Processing timing data…

core         WARNING     Fixed incorrect tyre stint information for driver '4'

req             INFO     Using cached data for car_data

req             INFO     Using cached data for position_data

req             INFO     Using cached data for weather_data

req             INFO     Using cached data for race_control_messages

core            INFO     Finished loading data for 20 drivers: ['4', '81', '1',
'63', '22', '23', '16', '44', '10', '55', '6', '14', '18', '7', '5', '12', '27',
'30', '31', '87']

core            INFO     Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]

req             INFO     Using cached data for session_info
```

```
req             INFO     Using cached data for driver_info
req             INFO     Using cached data for session_status_data
req             INFO     Using cached data for track_status_data
req             INFO     Using cached data for _extended_timing_data
req             INFO     Using cached data for timing_app_data
core            INFO     Processing timing data…
req             INFO     Using cached data for car_data
req             INFO     Using cached data for position_data
req             INFO     Using cached data for weather_data
req             INFO     Using cached data for race_control_messages
core            INFO     Finished loading data for 20 drivers: ['81', '63', '4',
'1', '44', '16', '6', '12', '22', '23', '31', '27', '14', '18', '55', '10',
'87', '7', '5', '30']
core            INFO     Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]
req             INFO     Using cached data for session_info
req             INFO     Using cached data for driver_info
req             INFO     Using cached data for session_status_data
req             INFO     Using cached data for track_status_data
req             INFO     Using cached data for _extended_timing_data
req             INFO     Using cached data for timing_app_data
core            INFO     Processing timing data…
req             INFO     Using cached data for car_data
req             INFO     Using cached data for position_data
req             INFO     Using cached data for weather_data
req             INFO     Using cached data for race_control_messages
core            INFO     Finished loading data for 20 drivers: ['1', '4', '81',
'16', '63', '12', '6', '44', '23', '87', '10', '55', '14', '30', '22', '27',
'5', '31', '7', '18']
core            INFO     Loading data for Miami Grand Prix - Qualifying [v3.8.1]
req             INFO     Using cached data for session_info
req             INFO     Using cached data for driver_info
req             INFO     Using cached data for session_status_data
req             INFO     Using cached data for track_status_data
```

```
req            INFO     Using cached data for _extended_timing_data
req            INFO     Using cached data for timing_app_data
core           INFO     Processing timing data…
req            INFO     Using cached data for car_data
req            INFO     Using cached data for position_data
req            INFO     Using cached data for weather_data
req            INFO     Using cached data for race_control_messages
core           INFO     Finished loading data for 20 drivers: ['1', '4', '12',
'81', '63', '55', '23', '16', '31', '22', '6', '44', '5', '7', '30', '27', '14',
'10', '18', '87']
```

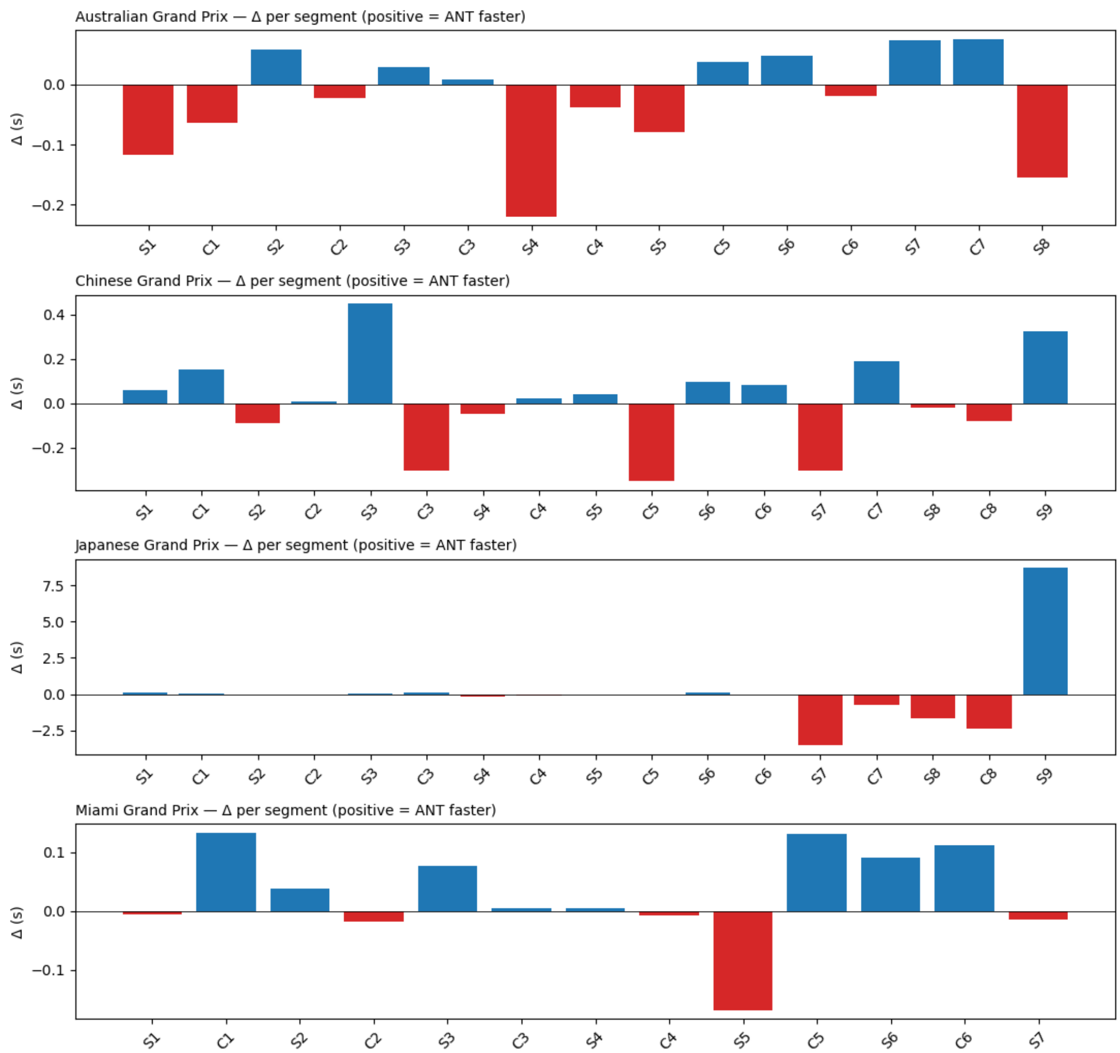
	Track	2025 Δ (s)	2026 Δ (s)	YoY gain (s)	2025 Q-mismatch?
0	Australia	-0.979	-0.293	0.686	True
1	China	-0.380	0.222	0.602	False
2	Japan	-0.237	0.298	0.535	False
3	Miami	0.114	0.399	0.285	False

```
Mean YoY gain across the 4 tracks: +0.527 s
```



1.9 Per-race detail (appendix)

For each race, the raw per-segment delta. Useful for curious readers; not the main story.



1.10 Caveats

- **Sample size: four races.** Findings are directional, not conclusive — and as the sensor investigation above shows, segment-level findings in particular are fragile to single-race data issues.
- **Q-session timing.** Q1 / Q2 / Q3 happen on an evolving track. The summary table flags any race where the two drivers' fastest valid laps came from different Q-segments.
- **Setup divergence.** Mercedes drivers do not always run identical setups. Public telemetry can't separate driver from setup.
- **Traffic and tire age within Q.** Out-laps, in-laps, and intra-session tire age all affect achievable lap time.
- **Telemetry sensor freezes.** FastF1's car telemetry occasionally stops reporting changes for long stretches. In this dataset, five Japan segments (S7, C7, S8, C8, S9 — covering D 4000 m to the end of the lap) are excluded by the sensor-quality filter above. Lap-level and sector-level deltas are unaffected — those come from timing-line beams, independent of the

car telemetry. Expect mild visual distortion at Japan's T14–T18 in the track-delta map.

1.11 What I found

The strongest finding in this project is the **year-over-year comparison**. At each of the same 4 tracks, Antonelli has gained between 0.3 and 0.7 seconds on Russell relative to his 2025 rookie season (mean **+0.53 s/track**). He started his rookie year almost a full second behind Russell at Australia and took until Miami (R6) to first cross ahead; in 2026 he started at −0.29 s and was ahead from R2 onwards. The rookie-year closing arc is being compressed, not just sustained.

The within-2026 trajectory — Antonelli being faster every round from R2 onwards, mean +0.16 s — is the second-strongest finding, and sector-time data confirms each lap-level delta independently of car telemetry.

The segment-level breakdown turned out to be much quieter than I first thought. My initial pass reported a +0.17 s/lap straights advantage and a −0.46 s/lap medium-corner deficit, but both numbers were almost entirely driven by five Japan segments where Antonelli's speed sensor was frozen near 189 kph for over 1.3 km of the lap. Once those segments are excluded, every category collapses to 0 s/lap (±0.01). At n=4 races and with one of them partially compromised, the segment-level story I went looking for didn't survive the data-quality check — which is worth noting, even if it wasn't the finding I initially wanted.

I'll re-run the same framework after each 2026 race — see What I'd build next for the extensions.